

WEEK 3— BUSINESS UNDERSTANDING REPORT

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Model Development & Performance Evaluation

Machine Learning Internship Programme

AnalystLab Africa

Week 3 Assignment

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1. Executive Summary

Customer churn is an important business problem because losing existing customers can negatively affect revenue, customer lifetime value, and long-term business growth. This project develops machine learning models capable of predicting customers who are likely to churn based on historical customer information.

Following the data understanding and preprocessing activities completed in Weeks 1 and 2, four supervised classification algorithms were developed and evaluated: **Logistic Regression, Decision Tree, Random Forest, and XGBoost**.

The models were evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, and Confusion Matrices. Among the models tested, **XGBoost achieved the strongest overall performance**, recording an accuracy of **80.27%** and ROC-AUC of **84.46%**.

The analysis also identified important predictors of churn, particularly **contract type, customer tenure, internet service type, online security, technical support, and payment method**.

Based on the results, XGBoost is recommended as the primary candidate for further validation and potential deployment as a customer retention decision-support system.

2. Business Problem

Customer churn occurs when customers discontinue their relationship with a company or stop using its services.

For a telecommunications company, high customer churn can result in:

- Loss of recurring revenue
- Increased customer acquisition costs
- Reduced customer lifetime value
- Difficulty maintaining a stable customer base
- Increased pressure on marketing and retention teams

The business therefore needs a way to identify customers who are **likely to churn before they leave**.

Machine learning provides an opportunity to analyze historical customer information and identify patterns associated with churn.

Business Objective

The primary objective of this project is to develop a classification model that can predict whether a customer is likely to churn.

The predictions can then support proactive customer retention activities.

3. Dataset and Modelling Data

The project uses the **Telco Customer Churn dataset** used throughout Weeks 1 and 2.

The original dataset contains:

7,043 customer records and 21 variables.

The preprocessing performed in Week 2 included:

- Converting TotalCharges to a numerical format
- Handling missing values
- Checking duplicate records
- Removing customerID
- Encoding categorical variables
- Scaling numerical variables
- Creating tenure_group
- Creating total_services
- Preparing the final machine-learning dataset

The final modelling data contained **50 processed features**.

Train/Test Split

Dataset	Records	Features
Training	5,634	50
Testing	1,409	50

The dataset was divided into approximately **80% training data and 20% testing data**.

The training set was used to develop the models, while the testing set was kept separate for evaluating their performance on unseen data.

4. Target Variable

The target variable is:

Churn

It contains two classes:

Churn	Meaning	Count	Percentage
0	Customer did not churn	5,174	73.46%
1	Customer churned	1,869	26.54%

The dataset therefore contains a class imbalance, with significantly more customers who did not churn than customers who churned.

This makes metrics such as **Recall, F1 Score, and ROC-AUC** important alongside accuracy.

5. Models Developed

Four classification algorithms were selected.

6.1 Logistic Regression

Logistic Regression provides a simple and interpretable baseline for binary classification.

It is useful for understanding whether a linear relationship between customer characteristics and churn can provide reasonable predictive performance.

6.2 Decision Tree

Decision Tree creates a series of decision rules based on customer features.

Its major advantage is interpretability because the decision-making process can be represented as a series of understandable conditions.

6.3 Random Forest

Random Forest combines multiple decision trees to produce a more robust prediction.

It can capture nonlinear relationships and interactions between customer characteristics.

6.4 XGBoost

XGBoost is a gradient boosting algorithm that builds models sequentially to improve predictive performance.

It was included because of its strong performance on structured/tabular datasets and its ability to capture complex relationships between features.

6. Model Performance

The models produced the following results:

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.799148	0.653199	0.518717	0.578241	0.842419
1	Decision Tree	0.798439	0.634731	0.566845	0.598870	0.829844
2	Random Forest	0.777147	0.602740	0.470588	0.528529	0.819997
3	XGBoost	0.802697	0.661074	0.526738	0.586310	0.844630

7. Model Comparison

The results demonstrate that the four models performed relatively well, but their strengths differed.

XGBoost achieved the highest accuracy, precision, and ROC-AUC.

Decision Tree achieved the highest recall and F1 score, meaning it identified a larger proportion of actual churners at the selected classification threshold.

Logistic Regression performed strongly and produced a ROC-AUC of 84.24%, making it a competitive and relatively interpretable baseline.

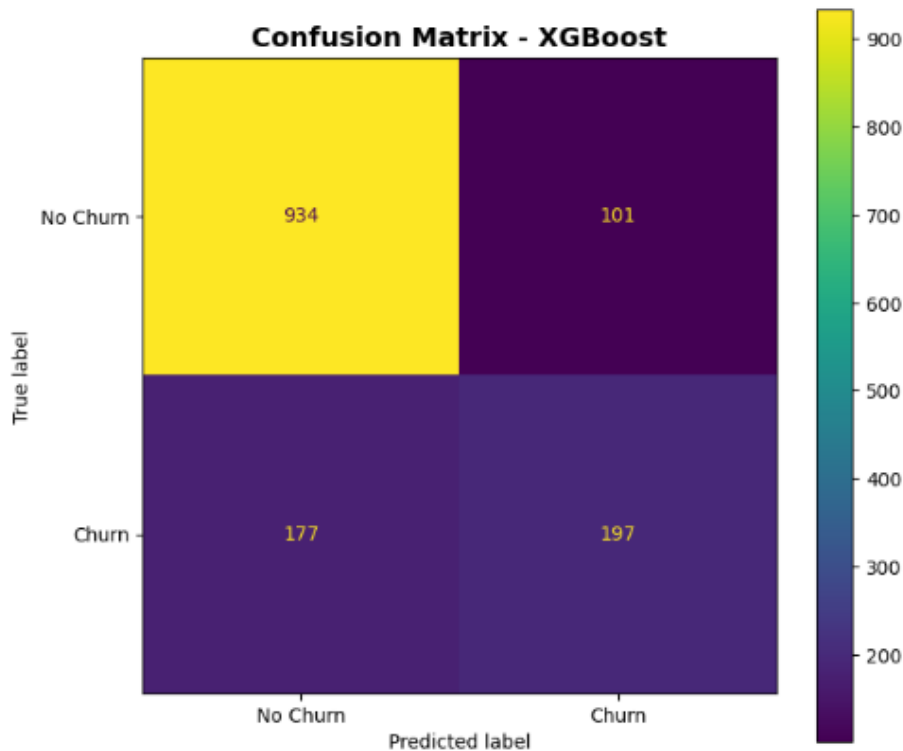
Random Forest produced the lowest overall performance among the four models in this experiment.

From an overall predictive perspective, XGBoost provides the strongest balance of performance.

8. Confusion Matrix Analysis

The confusion matrices provide additional insight into how the models classified churn and non-churn customers.

XGBoost



Therefore:

- **934** customers were correctly classified as non-churn.
- **197** customers were correctly classified as churners.
- **101** customers were incorrectly classified as churners.
- **177** actual churners were missed by the model.

The false-negative result is particularly important from a business perspective.

A false negative represents a customer who eventually churns but was not identified as a high-risk customer by the model.

Therefore, future optimization could focus on improving recall if the business considers missing potential churners more costly than contacting some customers who ultimately remain.

9. Feature Importance

The XGBoost model identified several important predictors.

The strongest features included:

	Feature	Importance
37	cat_Contract_Month-to-month	0.334742
49	cat_tenure_group_New	0.070030
17	cat_InternetService_Fiber optic	0.069545
19	cat_OnlineSecurity_No	0.065245
39	cat_Contract_Two year	0.045047
28	cat_TechSupport_No	0.039910
16	cat_InternetService_DSL	0.031238
38	cat_Contract_One year	0.024062
44	cat_PaymentMethod_Electronic check	0.019540
36	cat_StreamingMovies_Yes	0.019450
41	cat_PaperlessBilling_Yes	0.016822
1	num_tenure	0.016397
30	cat_TechSupport_Yes	0.015728
18	cat_InternetService_No	0.013337
40	cat_PaperlessBilling_No	0.011065
15	cat_MultipleLines_Yes	0.010387
13	cat_MultipleLines_No	0.010366
2	num_MonthlyCharges	0.009698
3	num_TotalCharges	0.009391
11	cat_PhoneService_No	0.009336

Business Interpretation

The very high importance of **month-to-month contracts** suggests that contract structure is strongly associated with churn prediction.

Customers with shorter tenure also appear important, suggesting that **new customers may require greater attention during the early stages of their relationship with the company.**

Other important factors include internet service type, online security, technical support, and payment method.

10. Recommended Model

Recommended Model: XGBoost

XGBoost is recommended as the primary candidate for deployment because it achieved:

- **80.27% Accuracy**
- **66.11% Precision**
- **52.67% Recall**
- **58.63% F1 Score**
- **84.46% ROC-AUC**

Most importantly, it achieved the **highest ROC-AUC and accuracy** among the evaluated models.

However, deployment should not occur solely based on these test results.

The model should first undergo:

1. Additional validation using new data.
2. Classification-threshold optimization.
3. Business cost analysis of false positives and false negatives.
4. Model monitoring after deployment.
5. Periodic retraining.

11. Business Recommendations

Based on the analysis, the organization should consider implementing a **proactive customer retention system**.

The system could assign each customer a churn-risk score.

Customers with high predicted risk could then be prioritized by the retention team.

Potential interventions include:

1. Focus on month-to-month customers

Provide incentives encouraging customers to move toward longer-term contracts.

2. Strengthen new-customer engagement

Customers in their early tenure could receive additional onboarding and support.

3. Improve technical support

Customers without technical support may represent an important retention opportunity.

4. Develop targeted retention campaigns

Instead of contacting every customer, the company can prioritize customers identified as high-risk.

5. Continuously monitor model performance

Customer behavior can change over time, so the model should be evaluated and retrained periodically.

12. Limitations

Several limitations should be considered.

First, the model is trained on historical data, meaning its predictions depend on patterns present in the dataset.

Second, the dataset contains class imbalance, with non-churn customers representing approximately 73% of observations.

Third, the model does not automatically explain the underlying cause of every individual customer's churn.

Finally, the current results represent an experimental evaluation and should be validated with new real-world customer data before deployment.

13. Next Steps

Future development should focus on:

- Hyperparameter tuning

- Threshold optimization
- Cross-validation
- Testing additional algorithms
- Addressing class imbalance
- Model explainability using SHAP or similar techniques
- Testing on new customer data
- Building a churn-risk dashboard
- Integrating predictions into customer retention workflows
- Monitoring model drift after deployment

Conclusion

This project demonstrated the complete transition from a prepared dataset to supervised machine learning models for customer churn prediction.

Four classification algorithms were developed and evaluated using multiple performance metrics.

Although the Decision Tree achieved the highest recall and F1 score, **XGBoost produced the strongest overall performance**, achieving **80.27% accuracy and 84.46% ROC-AUC**.

The analysis also revealed important churn-related predictors, particularly contract type and customer tenure.

The results demonstrate that machine learning can provide valuable decision-support capabilities for customer retention. With further validation, optimization, and responsible deployment, the recommended model could help organizations identify customers at risk of churn and take proactive action before customer attrition occurs.